

Table 1: Eight-dataset P0 generated-sample comparison under the locked short-budget protocols. Lower Frechet-style metrics are better.

Dataset	C low	D low	DDPM low	EDM low	low winner	C Inc	D Inc	DDPM Inc	EDM Inc	Inc winner	C rank
CIFAR-10	6.47	5.80	7.86	2.87	EDM	312.00	359.50	498.65	351.17	CoFiTok	1
Tiny-IN	8.49	6.99	9.07	3.10	EDM	373.63	392.40	381.24	268.72	EDM	2
ImageNet-64 HF	8.09	6.89	8.88	3.23	EDM	390.99	417.79	378.33	258.40	EDM	3
ImageNet-64 strict	8.36	6.87	8.80	2.75	EDM	374.68	383.72	355.11	248.31	EDM	3
FFHQ-64	8.74	6.84	9.44	2.51	EDM	425.71	427.05	369.01	325.49	EDM	3
AFHQv2-64	7.08	5.68	8.29	1.96	EDM	331.06	333.42	335.18	230.40	EDM	2
ImageNet-256 10%	10.44	12.59	10.40	10.39	EDM	414.19	366.42	419.60	368.68	Direct dense epsilon	3
ImageNet-256	10.54	10.24	10.40	10.40	Direct dense epsilon	417.26	354.13	414.30	368.43	Direct dense epsilon	4

C = CoFiTok; D = parameter-matched direct dense epsilon. Rows use the locked per-dataset short-budget protocol; they are not formal SOTA FID claims.

Table 2: Diagnostics for the supported CoFiTok claim.

Dataset	C PSNR	D PSNR	mask PSNR	C path	E path	C effK	E effK	C zero	deep-S zero
CIFAR-10	14.245	14.407	13.063	0.0967	0.7906	6.84	5.24	0.0000	0.0619
Tiny-IN	14.348	14.453	13.015	0.0731	1.1177	6.80	5.57	0.0000	0.0505
ImageNet-64 HF	14.897	15.004	13.548	0.0672	1.1120	6.79	5.33	0.0000	0.0522
ImageNet-64 strict	14.521	14.656	13.250	0.0705	1.0502	6.82	5.36	0.0000	0.0535
FFHQ-64	15.024	15.126	13.285	0.0676	1.0830	6.82	5.41	0.0000	0.0518
AFHQv2-64	15.167	15.154	13.384	0.0653	1.0185	6.82	5.45	0.0000	0.0561
ImageNet-256 10%	13.627	13.490	13.807	0.1416	1.5686	3.36	3.67	0.0000	0.0179
ImageNet-256	13.809	13.770	13.761	0.1325	1.5003	3.37	3.67	0.0000	0.0185

Lower path AUC indicates better prefix controllability. Zero-token ratio diagnoses whether the synthesis operator leaks structure at zero input.

Table 3: ImageNet-256 K4 20k confirmatory order and endpoint results over 1,024 validation images at $t = 500$.

Seed	E path	C ordered	non-ID mean	reverse	rank/24	Δ CI low	D endpoint	C endpoint
103	1.7879	0.0664	0.3157	0.7192	1	0.2489	0.1234	0.1133
139	1.4401	0.0678	0.3082	0.6821	1	0.2400	0.1244	0.1179

E = endpoint-only factorized; C = CoFiTok; D = parameter-matched direct dense. non-ID averages the other 23 K4 permutations; Δ CI low is the 95% paired-bootstrap lower bound for non-ID minus ordered path AUC.

Table 4: Official-pretrained tokenizer reconstruction context (eval-only).

Dataset	src	eval	FlexTok PSNR	FlexTok low	TiTok PSNR	TiTok low
AFHQv2-64	64	256	24.04	0.036	17.29	0.217
CIFAR-10	32	256	26.77	0.037	19.35	0.168
ImageNet-64 strict	64	256	22.57	0.042	16.73	0.196
FFHQ-64	64	256	24.06	0.042	17.39	0.231
ImageNet-64 HF	64	256	23.60	0.039	18.04	0.182
ImageNet-256	256	256	19.12	0.044	15.54	0.196
ImageNet-256 10%	256	256	19.12	0.044	15.54	0.196
Tiny-IN	64	256	21.81	0.044	17.30	0.160

Official pretrained reconstruction at 256x256 on 256 images per dataset. These rows are not retrained generation baselines.

Table 5: Official ImageNet-256 related-method evaluations (50K samples).

Method	Dataset	Protocol	Samples	FID	sFID	IS	Prec.	Rec.
D-AR	ImageNet-256	official 50K eval-only	50000	2.628	6.720	285.89	0.803	0.588
MAR	ImageNet-256	official 50K eval-only	50000	2.338	4.700	62.90	0.822	0.572
ReTok	ImageNet-256	official 50K eval-only	50000	2.219	5.896	245.94	0.816	0.599

These rows use official pretrained protocols where available and are secondary comparisons; they are not part of the matched-dataset/optimizer-step P0 matrix.

Table 6: Claim-evidence boundary.

Claim axis	Evidence summary
Generation quality	CoFiTok is best on 0/8 lowres rows and 1/8 Inception rows; average ranks are 3.12 and 2.62. Do not claim generation SOTA.
Prefix control	CoFiTok path AUC is lower than the endpoint-only factorized control on 8/8 datasets.
Repeated 20k evidence	CoFiTok lowers path AUC in 4/4 paired dataset-seed runs (mean relative reduction 96.18%), while endpoint x_0 MSE at $t = 500$ changes by +4.20% on average.
ImageNet-256 scaling	ImageNet-256 K4 20k confirmatory gates: 7/7 passed; overall PASS; mean endpoint-MSE change -6.73%; ordered ranks among all 24 permutations 1/1.
Restricted synthesis	CoFiTok zero-token ratio is zero in current rows; deep- S_k has nonzero zero-token ratio on 8/8 datasets.
Related methods	completed official 50K: D-AR, MAR, ReTok. All rows remain secondary and outside P0 matched-dataset/step training.
Main paper stance	The core empirical gates support the scoped claim of ordered restricted dense-noise factorization with prefix-controllable denoising, not broad unconditional generation superiority.